

Day 5 : Assemble and disassemble a PC Practical

Task 1 : List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Main Components Inside a Desktop PC

1. **Motherboard** – The main circuit board of the PC. It connects the CPU, RAM, storage, GPU, and other components so they can work together.
2. **CPU (Processor)** – The main processing unit or “brain” of the computer. It executes instructions and performs calculations.
3. **RAM (Memory)** – Provides temporary storage for programs and data currently being used. More RAM helps the computer handle multiple tasks smoothly.
4. **GPU (Graphics Processing Unit)** – Handles graphics, images, and videos. A dedicated GPU is useful for gaming, video editing, and other graphics-intensive tasks.
5. **HDD / SSD (Storage)** – Stores the operating system, applications, games, documents, and other files permanently. SSDs are generally faster than HDDs.
6. **SMPS / PSU (Power Supply Unit)** – Converts electricity from the wall socket into the correct power needed by the PC components.
7. **CPU Cooler** – Usually includes a heat sink and fan. It removes heat from the CPU and prevents it from overheating.
8. **Cabinet Fans** – Move air inside the cabinet to remove hot air and bring in cooler air. They help maintain proper airflow.

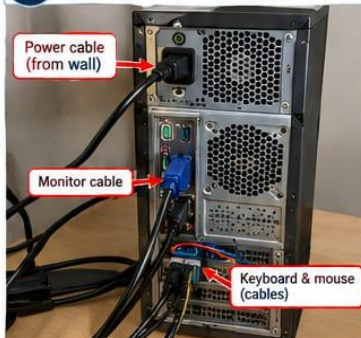
9. **CMOS Battery** – A small battery on the motherboard that keeps the system clock and certain BIOS/UEFI settings active when the PC is turned off.
10. **Optical Drive** – Used to read or write CDs and DVDs. It is optional and is less common in modern desktop PCs.
11. **Expansion Cards** – Additional cards that can be installed on the motherboard to add features such as Wi-Fi, sound, or extra connectivity.
12. **Cables and Connectors** – Carry power and data between different components, such as the motherboard, storage drives, GPU, and power supply.

Task 2 : Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Desktop PC Disassembly – Step by Step

Here are the steps I followed to disassemble a desktop PC, with clear photos and captions for each step.

1 Unplug Power & Peripherals



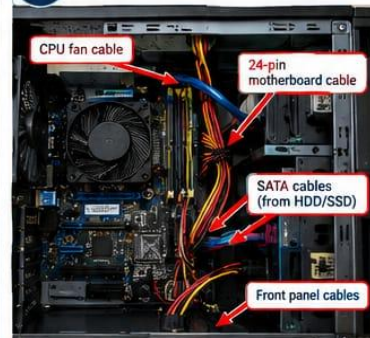
I turned off the PC, unplugged the power cable from the wall, and disconnected all peripheral cables (monitor, keyboard, mouse, etc.).

2 Remove Side Panel



I removed the screws from the back of the case and slid off the side panel to access the internal components.

3 Disconnect Internal Cables



I disconnected all the internal cables, including the 24-pin motherboard power cable, CPU fan cable, SATA data/power cables, and front panel cables.

4 Remove RAM



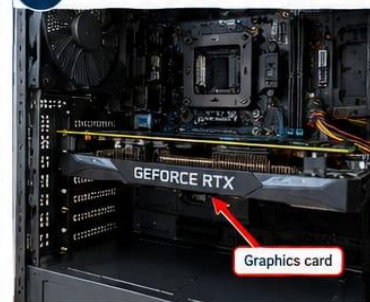
I pressed the side clips on the RAM slots and carefully removed the RAM sticks.

5 Remove Storage (HDD/SSD)



I removed the SATA cables and power cables from the HDD/SSD and unscrewed the drive from the case (if needed).

6 Remove Graphics Card (if any)



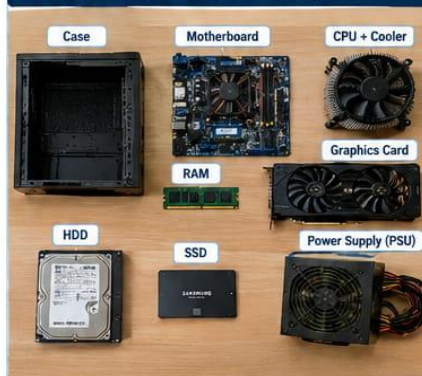
I unscrewed the bracket screws and released the locking clip on the PCIe slot, then carefully removed the graphics card.

7 Remove Motherboard



I removed all remaining screws, including the ones holding the motherboard, and carefully lifted it out of the case.

All Components After Disassembly



! Safety Tips

1. Always unplug the power cable and press the power button (with the PC off) to discharge remaining power.
2. Use an anti-static wrist strap (if available) to avoid damage from static electricity.
3. Keep screws and small parts in a safe place.
4. Work on a clean, well-lit surface.
5. Handle components carefully and avoid touching sensitive parts.

By following these steps, you can safely and properly disassemble a desktop PC.

Desktop PC Disassembly – Step-by-Step

1. Unplug Power & Peripherals

- Turn off the PC and unplug the power cable from the wall socket.
- Disconnect the monitor, keyboard, mouse, and other external cables.
- **Caption:** *I switched off the PC, unplugged the power cable, and disconnected all peripheral devices.*

2. Remove the Side Panel

- Remove the screws from the back of the computer case.
- Carefully slide the side panel off to access the internal components.
- **Caption:** *I removed the screws and carefully took off the side panel to access the inside of the PC.*

3. Disconnect Internal Cables

- Disconnect the power, SATA, fan, and front-panel cables connected to the components.
- Carefully remove each connector without pulling the wires forcefully.
- **Caption:** *I carefully disconnected all the internal power and data cables.*

4. Remove RAM

- Open the locking clips on both sides of the RAM slot.
- Hold the RAM module by its edges and gently remove it.
- **Caption:** *I released the RAM clips and carefully removed the RAM module from the motherboard.*

5. Remove Storage (HDD/SSD)

- Disconnect the SATA data cable and power cable from the storage drive.
- Remove the drive from its mounting area or bracket.

- **Caption:** *I disconnected the cables and carefully removed the HDD/SSD from the cabinet.*

6. Remove Graphics Card

- Remove the screw securing the graphics card to the case.
- Release the PCIe slot clip and carefully remove the graphics card.
- **Caption:** *I released the PCIe slot and carefully removed the graphics card.*

7. Remove Motherboard

- Disconnect any remaining cables and remove the motherboard mounting screws.
- Carefully lift the motherboard out of the computer case.
- **Caption:** *I removed the mounting screws and carefully lifted the motherboard out of the cabinet.*

Final Step – Completed Disassembly

- After removing all major components, I checked the cabinet to make sure nothing was still connected.
- I kept the motherboard, RAM, HDD/SSD, graphics card, cables, and screws safely organized for reassembly.
- **Caption:** *I completed the desktop PC disassembly and organized all components safely.*

Task 3 : Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

Reassemble the Desktop PC – Step-by-Step

1. Install the SMPS

- Place the SMPS in the cabinet and secure it with screws.
- **Why:** It supplies power to all the computer components.

Caption: *I installed and secured the SMPS inside the cabinet.*

2. Install the CPU

- Carefully place the CPU into the correct socket on the motherboard.
- **Why:** The CPU is the main processing unit of the computer.

Caption: *I installed the CPU into the motherboard socket.*

3. Install the CPU Cooler

- Place the cooler on the CPU and connect its fan cable to the motherboard.
- **Why:** It removes heat from the CPU and prevents overheating.

Caption: *I installed the CPU cooler and connected its fan cable.*

4. Install the Motherboard

- Place the motherboard on the cabinet standoffs and tighten the screws.
- **Why:** The motherboard connects and allows communication between all major components.

Caption: *I placed and secured the motherboard inside the cabinet.*

5. Install the RAM

- Insert the RAM into the correct slots until the clips lock.
- **Why:** RAM provides temporary memory for running programs and tasks.

Caption: *I installed the RAM modules into the motherboard slots.*

6. Install the HDD/SSD

- Place the storage drive in its bay or slot and connect the power and data cables.
- **Why:** It stores the operating system, applications, games, and personal files.

Caption: *I installed the HDD/SSD and connected the required cables.*

7. Install the GPU

- Insert the GPU into the PCIe slot and secure it with a screw. Connect its power cable if needed.
- **Why:** The GPU processes graphics for gaming and other graphics-intensive tasks.

Caption: *I installed and secured the GPU in the PCIe slot.*

8. Connect All Cables

- Connect the SMPS cables to the motherboard, CPU, GPU, and storage. Also connect the power button, USB, audio, and fan cables.
- **Why:** Proper connections provide power and allow all components and cabinet features to work.

Caption: *I connected the power, front-panel, fan, and other required cables.*

9. Final Check and Close the Cabinet

- Check that all components are properly installed and cables are secure. Then replace the side panel.
- **Why:** A final check helps prevent loose connections and startup problems.

Caption: *I checked all connections and closed the cabinet.*

10. Power On and Test

- Connect the external cables and turn on the PC. Check that the system detects the CPU, RAM, storage, and GPU.
- **Why:** This confirms that the PC has been correctly reassembled.

Caption: *I powered on the PC and checked that all major components were working properly.*

Task 4 : Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

PC Assembly/Disassembly – Safety Precautions and Tools

A. Safety Precautions

1. **Turn Off and Unplug the PC** – Shut down the computer and remove the power cable before opening the cabinet. This prevents electric shock and component damage.
2. **Remove External Cables** – Disconnect the monitor, keyboard, mouse, and other cables to make the PC easier and safer to handle.
3. **Prevent Static Electricity** – Use an anti-static wrist strap or touch a grounded metal surface before handling components. Static electricity can damage sensitive parts.
4. **Use a Clean and Dry Workspace** – Work on a flat, clean, and dry surface to prevent dust, moisture, and accidental damage.
5. **Handle Components Carefully** – Hold RAM, GPU, and motherboard by their edges. Avoid touching the pins, connectors, and circuits.
6. **Keep Screws Organized** – Place screws in a small container and remember where they belong. This makes reassembly easier.

B. Tools Needed

1. **Phillips Screwdriver** – Used to remove and tighten screws in the cabinet, motherboard, GPU, and other components.
2. **Anti-Static Wrist Strap** – Protects computer components from electrostatic discharge (ESD).
3. **Small Container** – Keeps screws and other small parts safely in one place.

4. **Compressed Air** – Helps remove dust from fans, heatsinks, and other internal components.
5. **Soft Cloth** – Useful for cleaning dust from the cabinet and components.
6. **Flashlight** – Helps identify small screws, connectors, and cables inside the cabinet.